Arabian Journal of Geosciences (2018) 11: 446
https://doi.org/10.1007/s12517-018-3806-x

ORIGINAL PAPER

Equatorial paleolatitude for Northeast Africa in the Late Triassic: paleomagnetic study on the Gezira and Bir-Umhebal [229–223 Ma] ring complexes, Southeastern Desert, Egypt

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Received: 3 February 2017 / Accepted: 7 August 2018 / Published online: 16 August 2018
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Abstract

Two Late Triassic alkaline ring complexes in the Southeastern Desert of Egypt: the Gezira [229 ± 5 Ma, K/Ar biotite] and Bir-Umhebal [223 ± 9 Ma, Rb/Sr], were paleomagnetically studied to determine the paleo-latitude position of Northeast Africa during the Late Triassic just before the Jurassic dispersion of Pangaea. The initial isothermal remanent magnetization [IRM] study was followed by the progressive thermal demagnetization of all samples to verify the characteristic natural remanent magnetic directions of both complexes. The analysis revealed that: (i) the magnetite-dominated sites generally, had dual-polarity, bi-vectorial with shallow sub-equatorial N–S to NNW–SSE anchored remanent magnetic component, while (ii) the hematite/goethite sites were characterized by normal polarity, bi-vectorial decay with a northward anchored component fluctuating around zero inclination.

Based on their thermal demagnetization behavior, only the magnetite-dominated sites were considered as representing the average Late Triassic paleomagnetic field in the study areas during cooling of the studied complexes. The mean paleomagnetic north poles of the isolated characteristic component using the principal component analysis [PCA] for each complex were: (i) the north poles of Gezira Complex [22.3°N/33.6°E] was at (54.6°N/226.2°E, Dp/Dm= 5°/9°); (ii) the north poles of Bir-Umhebal Complex [23.75°N/28 33.3°E] was at (56.4°N/236°E, Dp/Dm= 3°/6°), and (iii) the mean north pole of both complexes [23°N/33.5°E] was at (56°N/231°E, Dp/Dm = 3°/6°).

The obtained Late Triassic [Late Carnian-Early Norian] paleomagnetic poles for the two ring complexes of Egypt show reasonable consistency with their synchronous African poles as well as poles rotated from the main tectonic units to the Northeast Africa using appropriate Euler pole parameters. These poles reveal that in the Late Triassic:The obtained Late Triassic [Late Carnian-Early Norian] paleomagnetic poles for the two ring complexes of Egypt show reasonable consistency with their synchronous African poles as well as poles rotated fromthe main tectonic units to the Northeast Africa using appropriate Euler pole parameters. These poles reveal that in the Late Triassic: (i) the northern part of Africa was in Equatorial paleo-latitudes with Northeast Africa lying just south the Equator. Cairo [Now at 30°N] was at −4.4° to −1° paleo-latitude; (ii) the Azimuth of the African Plate was between N and 37 NNE about 9° to14° clockwise to its present-day azimuth.

Keywords Paleo-latitude · Paleomagnetism · Paleo-tectonic position of Africa · Egypt · Triassic · Ring complex

Introduction

After the Late Paleozoic amalgamation of Pangea, the Neotethys started rifting in the Late Permian (Bernoulli et al. 1990), then growing on the expense of the northward subducting Paleotethys during the Triassic (Sengör et al. 1988). Despite this, the Late Triassic witnessed the commencement of the Neotethys sea-floor spreading and rifting of the northern Gondwana margin, while the continental separation did not occur before the Early Jurassic in the Egyptian segment in northeast Africa (Schandelmeier et al. 1997).

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Within north Gondwana, the extensional tectonics associated with the Triassic evolution of the Neotethys created a collage of intercontinental small ocean basins separated by micro-continents in the Eastern Mediterranean (Robertson et al. 1991). This extension was diachronous affecting Libya and Tunisia in Early to Middle Triassic, the Levant in Middle to Late Triassic (Mart 1987), then it reached northeast Africa and Egypt in Late Triassic (Keeley 1994).

The magmatism related of the Late Triassic extensional tectonics in northeast Africa is documented by a peak of intra-plate magmatic pulses spreading in North Sudan and South Egypt in the form of numerous intrusive alkaline ring complexes and subvolcanic rocks. In south Egypt, Meneisy (1990) recognized a major Permo-Triassic magmatic pulse recorded in the Southeastern Egyptian Desert by Gezira ring complex [229 Ma] (Serencsits et al. 1979, 1981) and Bir-Umhebal complex [223 Ma] (Hashad and El-Reedy 1979) and in the southern Western Desert of Egypt at Bir Safsaf alkaline rhyolitic subvolcanics [216 Ma] (Schandelmeier and Darbyshire 1984).

Structurally, Garson and Krs (1976) referred the emplacement of the alkaline complexes in the Southeastern Desert to the intra-plate magmatic activity occurring along the intersection between two old regional fracture systems: the N60°E transform faults and the N30°W deep fracture zones.

The present study sheds light on the Late Triassic paleolatitude of North Africa, just before the dispersion Pangea through the rifting of the Central Atlantic Ocean and the drifting of Gondwana away from Laurasia in the Early Jurassic.

Geology

The Gezira ring complex [22.3°N/33.6°E] has a well-defined ring structure measuring about 3.5 km across with an elevation of 483 ms. It protrudes through the Precambrian igneous and metamorphic complex of the Southeastern Desert of Egypt (Fig. 1) and exhibits high-degree of rock differentiation represented by a wide spectrum of alkaline rocks ranging from trachybasalts to syenites (El Ramly and Hussein 1982, 1985). According to El Ramly et al. (1970, 1971), the Gezira ring complex (Fig. 1c) is made up of a central core mass surrounded by two rings separated by two crescent-shaped valleys. The outer ring seems to be the relics of cone volcanics and flows of trachybasalts, trachytes and latites while the inner ring is made up of alkaline syenites. Essexite gabbroic rocks are intruded between the two rings and the central mass. The central mass of the complex exposes mainly hypabyssal alkaline syenite intruded through the volcanic flows with a stock-like body of pegmatoidal alkaline syenite occupying the northeastern part of the complex. Several microsyente dykes and veins intrude the complex.

According to El Ramly et al. (1970, 1971), the formation of the ring complex started with the development of a volcanic cone encircled by successive flows of trachybasalt to latites and rhyolites which form now the outer ring of the complex. These flows were, then, intruded by an incomplete ring of essexite gabbro followed by the alkaline syenite forming the inner ring. At last the alkaline syenite central mass was intruded followed by the dykes.

The age of Gezira ring complex was determined by Serencsits et al. (1979, 1981) using K/Ar technique on biotite from the gabbro as 229 ± 5 Ma (Meneisy 1990). This age is Late Carnian or Early Late Triassic according to Cohen et al. (2013), Gradstein et al. (2012) and Ogg et al. (2016) geologic time scales.

Bir-Umhebal Ring Complex [23.25°N/33.3°E] lies about 60 kms southeast of the monumental city of Aswan (Fig. 1). It is made up of a complete ring dyke of alkaline syenite, microsyenite and nordmarkite (about 3 kms across) intruding and enclosing older cone volcanics of trachybasalt, trachyte and tuffs (Fig. 1b). The whole ring protrudes through the Precambrian porphyritic Aswan monumental red granite which extends southeastwards of Aswan province (Helmy et al. 1979) at Bir-Umhebal water well.

Extensive trachybasalt and trachyte flows, frequently, cut by deep crescent-shaped valleys extend more than 14 kms northwestwards out of the ring complex. These lava flows cover the Cretaceous Nubia- type sandstone to the east of the granite.

The age of Bir-Umhebal ring complex was defined by Hashad and El-Reedy (1979) using Rb/Sr (WR) of the syenite ring as 223 ± 9 Ma (Meneisy 1990). This age is Early Norian or Middle Late Triassic according to Cohen et al. (2013), Gradstein et al. (2012) and Ogg et al. (2016) geologic time scales.

Sampling

A number of oriented rock samples were collected from 21 sites representing all rock-units and varieties in the Gezira ring complex (Fig. 1c). Samples were collected from different elevations along the periphery of the outer ring of this complex, from the inner ring and from the central mass following the crescent-shaped wadies separating the rings and the central mass.

On the other hand, oriented samples were, also, collected from 22 sites allover Bir-Umhebal ring complex (Fig. 1b). Sampling was concentrated on the main ring structure covering both the inner volcanic cone basalt, trachybasalt and tuffs as well as the outer intrusive ring dyke alkaline syenite, microsyenite and nordmarkite. About half of the sites were spread over the trachybasalt/basalt flows

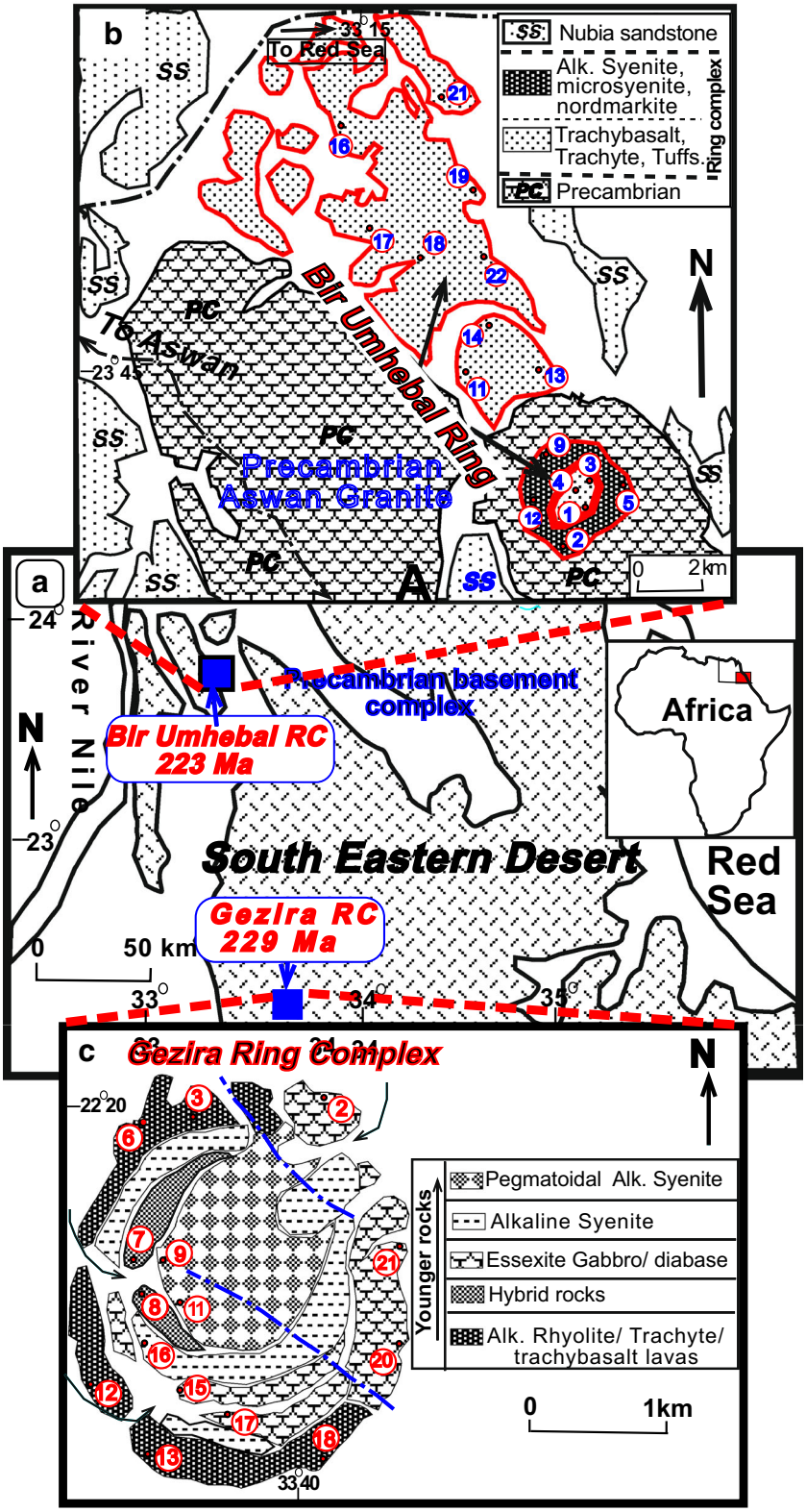


Fig.1 **a** Location map for the studied ring complexes in the Southeastern Desert of Egypt. **b** Modified geologic map for Bir-Umhebal ring with the surrounding area (Unanimous 1996). **c** Simplified geologic map for Gezira ring complex after El Ramly and Hussein (1982, 1985). Circled numbers in **b** and **c** represent the location of the sampling sites

extending northwestwards from the ring complex. A number between 8 and 6 individually-oriented cores was maintained and collected at each site using a solar compass for orientation.

Isothermal remanent magnetization [IRM]

The paleomagnetic measurements, which, were carried on an Agico JR-6 dual-speed spinner magnetometer [sensitivity 2 ×

10^{-3} mA/m], housed in the paleomagnetic laboratory of the National Research Institute of Astronomy and Geophysics [NRIAG] in Egypt, included both the preliminary IRM study followed by the natural remanent progressive stepwise thermal demagnetization.

The rock magnetic study was applied as an initial step, before the paleomagnetic study, in order to classify the collected sites according to their magnetic mineral associations. Taking into consideration that the primary opaque minerals in igneous rocks are of the ilmenite-magnetite family, the initial IRM study was designated to verify their contribution in the natural magnetic remanence.

One sample from each site was subjected to the IRM study according to the following scheme. The IRM was progressively acquired up to 800 mT, starting with narrow steps (25 mT) up to 100 mT, to verify the magnetite saturation. The saturated samples were, then, rotated 180° in the electromagnet and exposed to back-field demagnetization, using 25 mT steps (sometimes narrower steps) down to −150 mT. The samples were, then, re-saturated up to 800 mT and progressively thermally demagnetized up to 680 °C. The thermal demagnetization started with 50 °C steps till 200 °C, to define the goethite constituents, then ended with 50 °C steps from 400 °C till 680 °C while approaching the Curie temperatures of the magnetite (585 °C) and hematite (680 °C).

The applied IRM measurements (Fig. 2) were capable to categorize four groups of sites according to their overall IRM behavior:

1. The magnetite- dominated (fresh) sites which represent most of the collected sites (Fig. 2). These sites were characterized by early sharp magnetic saturation < 50 mT, rapid back-field demagnetization <−50 mT and complete thermal demagnetization between 400 °C- 580 °C, reflecting magnetite as the sole carrier of remanent magnetization
2. The magnetite/hematite/goethite bearing sites, like site GZ-5-2 (Fig. 2), were initially partly saturated at <50 mT reflecting the presence of magnetite, but never reached saturation up till 800 mT due to the coexistence of hematite and/or goethite. The back-field demagnetization started by a sharp drop <−50 mT due to the magnetite but never demagnetized down to −150 mT due to hematite/goethite. Upon thermal demagnetization, these samples started by sharp drop in intensity <200 °C due to the goethite, followed by steady decrease till 680 °C pointing to coexistence of magnetite and hematite. These sites were discarded from the thermal demagnetization due to the expected complex behavior.
3. The Hematite/goethite bearing sites, like sites GZ-8-3 and Bir-21–2 (Fig. 2), were characterized by steady increase in magnetic intensity without reaching saturation up to 800 mT. The back-field demagnetization curve was steadily decaying without reaching complete demagnetization down to −150 mT due to the dominance of hematite and goethite. The thermal demagnetization path was dominated by two major drops in intensity; an initial drop <200 °C followed by another drop at >600 °C, without intermediate decay, reflecting the coexistence of goethite and hematite and absence of magnetite. In spite of the absence of magnetite in these sites, they proceeded to the paleomagnetic study in order to shed light on the magnetic remanence carried by the hematite after the removal the thermally soft goethite remanence.
4. The goethite-dominated sites, like Bir-8-2 (Fig. 2) were characterized by steady increase in intensity without reaching saturation up to 800 mT. Upon reversing the magnetic field, the samples were never demagnetized. The thermal demagnetization was characterized by a sharp complete drop and loss in intensity <200 °C, pointing to goethite as the sole remanence carrier. This group of sites was eliminated from the paleomagnetic study as they are expected to be overprinted by present-day field magnetization residing in the secondary goethite.

Thermal demagnetization of the natural remanence

The magnetite-dominated samples and the hematite/goethite samples were, all, exposed to progressive stepwise thermal demagnetization up to fifteen steps or till complete demagnetization. Demagnetization started by 50 °C increments till 200 °C to isolate the remanence carried by the goethite, followed by 100 °C steps till 400 °C. The 50 °C increment demagnetization was retained above 400 °C till 550 °C then 25 °C steps till complete demagnetization to verify the anchored magnetic component carried by magnetite and/or hematite in the Gezira and Bir-Umhebal ring complexes.

The inspection of the orthogonal projections (Zijderveld 1967) (Fig. 3), revealed that almost all samples were characterized by bi-vectorial decay. They started by the early decay of a soft northward normal medium inclination variable-length component removed at low temperature < 200 °C (Fig. 3a, b). This component is parallel to the present-day direction of the geomagnetic field [PDF] in the study area and is considered as a partial PDF overprint carried by goethite. Subsequently, the magnetic demagnetization behavior diverges:

1. The magnetite-dominated samples, which represent the majority of the samples, started a clear steady decay of a NNW-SSE component with a shallow antipodal inclination with anchored trajectories until

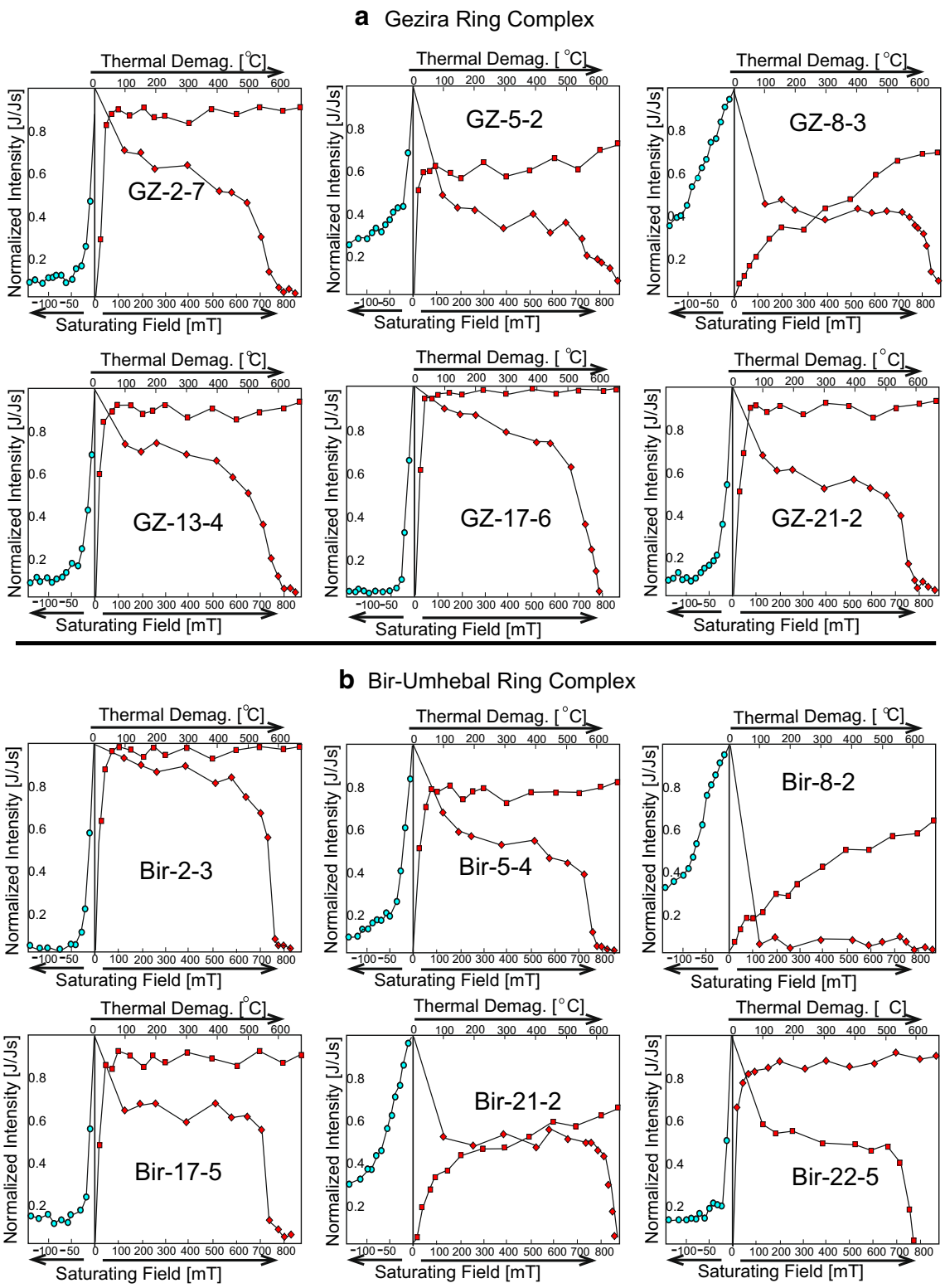


Fig. 2 Isothermal remanent magnetization [IRM] behavior of some samples from the studied ring complexes. IRM acquisition curves are the ascending lines joining the closed squares which represent the acquisition steps to the right of the vertical separator. The back-field demagnetization curves are the descending curves connecting the circle steps to the left of the vertical separator. The thermal demagnetization curves are the descending curves joining the closed diamond steps to the right of the vertical line

complete demagnetization at about 585 °C. This stable component which was diagnostic in most magnetite-dominated samples (Fig. 3a, b) in the studied ring complexes was considered as the

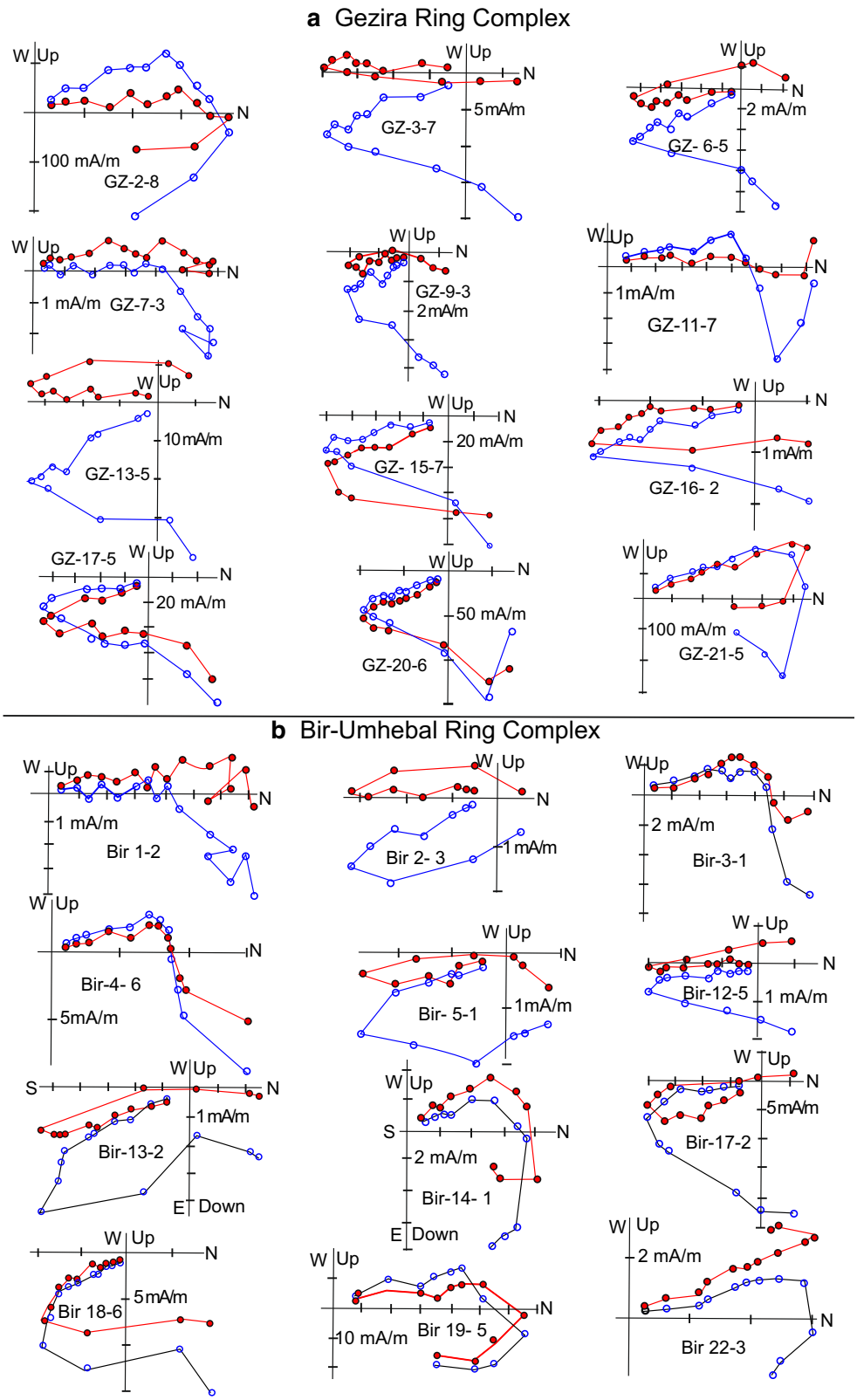


Fig. 3 Orthogonal projections (Zijderveld 1967) of the demagnetization curves during progressive thermal demagnetization. **a** Gezira R. C., **b** Bir-Umhebal R. C. In the orthogonal projections, open circles (filled circles) denote the vertical (horizontal) projections

characteristic remanent magnetization of both Gezira and Bir-Umhebal complexes representing the magnetic field during their formation in the Late Triassic [229–223 Ma].

2. The Hematite/goethite samples, on the other hand, after the removal of the goethite-bearing PDF overprint, started the decay of a normal northward component swinging around zero inclination till 680 °C. Based on its

Table 1 Paleomagnetic demagnetization results of Gezira and Bir-Umhebal ring complexes, N = number of demagnetized samples/site, n = number of isolated directions/site, PDF = present-day field overprint, ChRM = characteristic remanent magnetization, Min. = mineral carrying the magnetic component, Mg = Magnetite and Hm = Hematite, Dec. = declination [°], Inc.,=inclination [°], K = kappa precision parameter (Fisher 1953), α95 = semi-angle of the cone of 95% confidence [°] (Fisher 1953), VGP [°N/°E] = Virtual geomagnetic north pole locations, Dp/Dm = semi-angle of the cone of 95% confidence about the paleomagnetic pole in the co-latitude direction (Dp) and at right-angle (Dm). Sites marked with [*] are not included in the means because of being secondary and/or their wide dispersion

Demagnetized sites/samples			Soft component [PDF]			Anchored component [ChRM]				
Site	Rock type	*N*	*n*	Dec.°/ Inc.°	K/α95°	*n*	Min.	Dec.°/Inc.°	K/α95°	VGP [°N/°E]
Gezira alkaline ring complex [22.3°N/33.6°E] [229 ± 5 Ma]										
GZ-2	Essexite gabbro	8	5	14/34	46/11.4	8	Mg	354/−34	41/8.8	48.5/222.2
GZ-3	Alkaline Trachyte	8	4	3/27	42/14.4	8	Mg	186/24	38.8/9	54.7/203.4
GZ-6	Alkaline Trachyte	6	6	339/43	16/17.1	6	Mg	172/26	44.5/10	53.2/226.6
GZ-7*	Hybrid Rocks	4	4	359/48	45.3/13.8	4	Hm	351/−3	13/21	64.5/235
GZ-8*	Hybrid Rocks	4	3	18/48	37/15.3	3	Hm	349/−9	16/29	61/237
GZ-9*	Pegm. Alk. Syenite	4	4	10/35	27.8/11.7	3	Mg	154/31	16/24	43/249
GZ-11	Pegm. Alk. Syenite	7	3	4/58	62/15.8	7	Mg	358/−11	41/9.5	62/218
GZ-12*	Trachy-basalt lava	5	3	3/46	28/23.8	5	Hm	18/46	12/24	PDF
GZ-13	Trachy-basalt lava	5	4	342/19	23.8/19.2	5	Mg	184/39	34/13.2	45.5/208.3
GZ-15	Essexite Diabase	8	4	1/37	26/18.3	7	Mg	169/11	35/10.4	60/236
GZ-16	Alkaline Syenite	5	4	4/13	47.2/13.5	5	Mg	176/13	46/11.4	61/222
GZ-17	Essexite gabbro	6	6	19/25	36/11.3	6	Mg	164/13	47.7/9.8	57/244
GZ-18*	Trachy-basalt lava	5	5	9/44	31.4/13.9	4	Hm	3/−6	12/39	64.5/207
GZ-20	Essexite gabbro	8	7	29/57	42.8/9.3	8	Mg	159/35	44.7/8.4	43.6/241.5
GZ-21	Essexite gabbro	8	3	6/39	54.2/16.9	8	Mg	344/−28	43.6/8.5	52/240
	Mean of Gezira R.	15/ 91	15/ 69	4.5/38	18.5/10.9	10/ 68		352.6/−23.6	37/8.0	54.6/226.2 Dp = 5° Dm = 9°
Bir-Umhebal alkaline ring complex [23.75°N/33.3°E] [223 ± 9 Ma]										
Bir-1*	Trachyte	5	5	348/30	47/11.3	3	Hm	358/−8	12/26	63.5/218
Bir-2	Alkaline Syenite	6	3	335/17	75/14.4	5	Mg	184/16	49/11	58/206
Bir-3	Trachyte porphyry	7	4	7/35	22.8/19.7	6	Mg	348/−15	74/7.8	56.5/235
Bir-4	Trachyte porphyry.	7	6	19/47	28.3/12.8	7	Mg	345/−16	39.5/9.7	55/240
Bir-5	Microsyenite	4	4	22/41	42/14.3	4	Mg	171/16	54/12.6	57/230
Bir-9*	Nordmarkite	6	6	9/18	79.7/7.6	5	Hm	2/57	28/13	PDF
Bir-11*	Trachyte	5	5	5/37	51/10.8	3	Hm	357/−1	15/27	67/221
Bir-12	Alkaline syenite	7	5	356/24	34.7/13.2	7	Mg	180/6	35/10.4	63/213
Bir-13	Trachyte	7	4	3/22	29/17.4	6	Mg	161/21	47.6/9.8	51/243.6
Bir-14	Trachyte	7	3	19/41	81/13.8	7	Mg	332/−16	45.4/9.1	48/257
Bir-16*	Tuffs	4	4	346/50	57.7/12.2	4	Mg	278/11	9/22	intermediate
Bir-17	Trachy-basalt	6	2	356/51	—/—	6	Mg	164/7	38.8/11	58.6/245
Bir-18	Trachy-basalt	7	4	3/44	55.7/12.4	6	Mg	167/27	45.3/10	50/230
Bir-19	Trachy-basalt	5	5	20/17	33.3/13.4	5	Mg	351/−11	53/10.6	59/231
Bir-21*	Tuffs	3	3	6/40	27.5/14.8	3	Hm	339/−8	27/16	56.5/254
Bir-22	Trachyte	6	4	339/31	36.9/15.3	5	Mg	341/−11	62.4/9.8	55/248
	Mean of Bir-Umhebal R.,	16/ 86	16/ 67	2/35	24.4/7.7	11/ 64		347.6/−15	59.6/6	56.4/236 Dp = 3°/Dm = 6°
Mean of Gezira and Bir-Umhebal complexes *N* = 21 sites						21/ 132		350/−19	41/5	56/231 Dp = 3°/Dm = 5.6°

occurrence in hematite-dominated samples in both ring complexes like GZ-7-3 and Bir-1-2 (Fig. 3a, b), this stable anchored component was considered as a secondary chemical remagnetization residing in secondary hematite

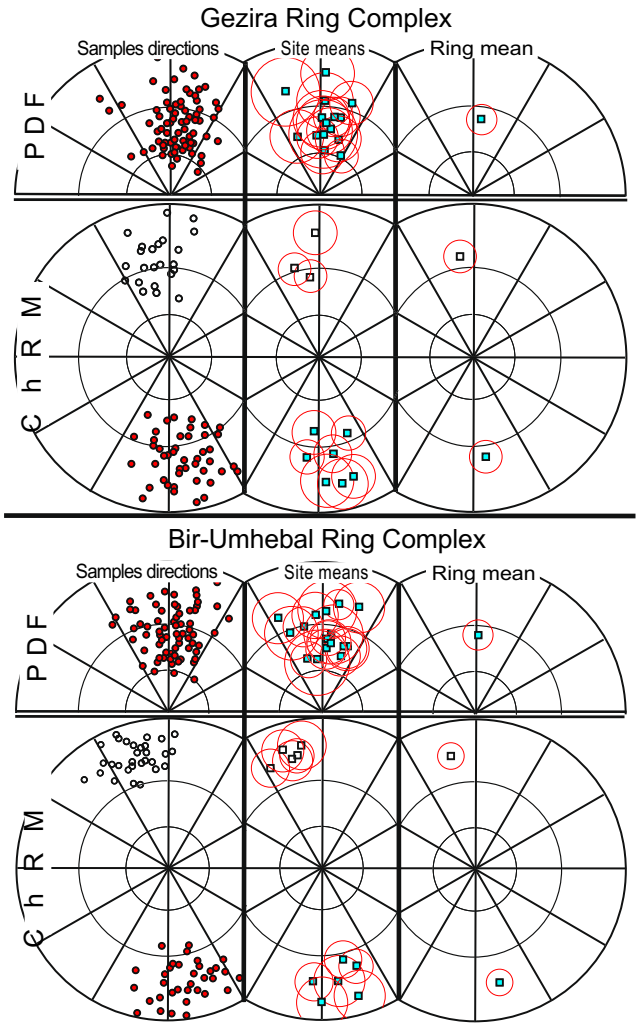


Fig. 4 Equal area projection showing the directions of the isolated demagnetized magnetic components in both Gezira and Bir-Umhebal Ring Complexes. In the left part, the directions of both the PDF overprint (upper) and the direction of the ChRM component (lower) of all samples. In the central part, the site-means with their semi-angles of the cones of 95% confidence of both components. In the right side, the respective overall ring-mean directions

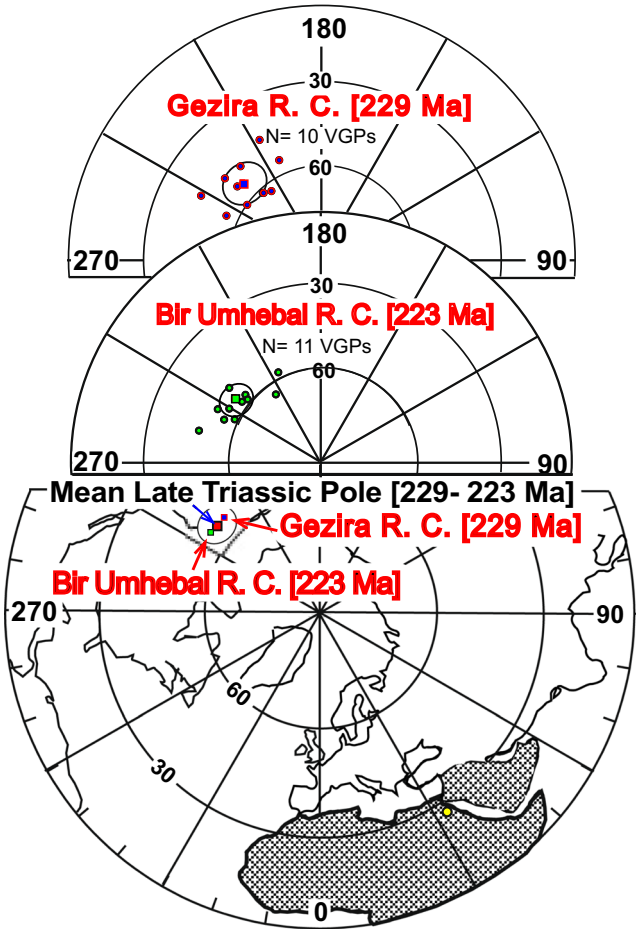


Fig. 5 Equal area projections for the virtual geomagnetic poles [VGPs] of the site-means of the ChRM together with the mean paleomagnetic pole of each ring complex, and the overall Late Triassic (229–223 Ma) paleomagnetic pole

reflecting a regional chemical remagnetization event in the Southeastern Desert of Egypt.

Paleomagnetic interpretation and paleomagnetic poles

The visual inspection of the decaying trajectories in the orthogonal projections of the demagnetized samples (Fig. 3), enabled the isolation of an early-decayed soft component (<200 °C) and two different high-temperature anchored components carried by either magnetite (450°–585 °C) or hematite (>585 °C). The soft component, which was north-oriented normal and medium inclination, dispersed around the present-day direction of the geomagnetic field [PDF] in the study area, is carried by goethite and considered as the PDF overprint in the samples. On the other hand, the magnetite component was antipodal N-S to NNW-SSE shallow inclination while the hematite component was normal-polarity fluctuating around the north direction and swinging around zero inclination.

For each sample in the two ring complexes, the best-fit line direction of at least three successive demagnetization steps, of both the soft overprint and the anchored components, was calculated using the principal component analysis technique (Kirschvink 1980). Unclear short (less than three steps) or curved components were ignored. Then, for each site, the site-mean direction of both the PDF and the diagnostic anchored components were calculated with their associated precision parameter (K) and semi-angle of the cone of 95% confidence about the mean ($\alpha 95$) (Fisher 1953) (Table 1).

It was noticed that the northward equatorial shallow inclination component characterizing the hematite-dominated sites, which is recorded in only four sites in each ring complex, show a high site-mean dispersion

Table 2 Euler pole rotation parameters used to rotate the paleomagnetic poles of the main tectonic units to NE Africa coordinates. Latitudes are reckoned (+) North, and (−) South, longitudes are (+) East, and (−) West, rotation angles are (+) counterclockwise and (−) clockwise

Tectonic units rotated	Euler pole rotation parameters			Reference
	Latitude[°]	Longitude[°]	Angle[°]	
Greenland versus Stable Europe	65.1	126.1	18.9	Alvey 2009
Stable Europe versus N. American Craton	78.6	161.9	-31	Alvey 2009
N. American Craton versus NW Africa	64.3	345.3	78	Labails et al. 2010
Colorado Plateau versus N. American Craton	35	−102	10.5	Beck and Housen 2003
Amazonia versus S. Africa	50.0	327.5	55.1	Torsvik et al. 2009
Colorado versus S. Africa	47.5	326.7	57.3	Torsvik et al. 2009
NW Africa versus S. Africa	33.6	26.0	2.3	Torsvik et al. 2012
S. Africa versus NE Africa	40.5	−61.4	0.7	Torsvik et al. 2012

Table 3 The Late Triassic Paleomagnetic North poles of the North American Craton rotated to NE Africa coordinates using Euler pole rotations parameters in Table 2. Colorado Plateau poles marked by (*) are internally rotated to the North American Craton using rotation parameters after Beck and Housen (2003) in Table 2, [O] original poles, [R] rotated poles. All poles are corrected for inclination error be rotation to Africa

Rock unit	Age [Ma]	α95	North Poles °N/°E	North Poles Corrected For Inc. error	NE Africa Coordinates °N/°E	Reference
Chinle Fm., Upper shale mb., NM	216	5.6	57.4/87.8	59.3/98.4	59/233.4	Bazard and Butler 1991
Newark Basin [Titusville core], NJ	217	3.2	56.2/99.9	60/99.5	58.7/235	Kent et al. 1995 Kent and Tauxe 2005
Dockum Gp, NW Texas	218	7.7	56.4/96.3	58/97.6	59/230.7	Molina-Garza et al. 1995
Chinle Fm., Sangre de Cristo Mts., NM*	218	5.1	50.8/109 [R] 54.3/92.6 [O]	51.6/110.2	49.5/227.3	Molina-Garza et al. 1996
Chinle Fm., Chinarump Mb., NM*	220	5	57.7/108.5 [R] 60.8/88.9 [O]	61.6/110.8	54.3/241.6	Molina-Garza et al. 1991
Newark Basin sed., PN	220	6	53.5/101.6	55.9/101.5	56.1/228.9	Witte and Kent 1989
Dan River Fm., Danville rift Basin. N Carolina, Virginia	221	2.5	50.4/100.1	58.5/99.8	58/232.5	Kent and Olsen 1997 Kent and Tauxe 2005
Newark Basin [Nursery core], NJ	221	2.6	54.1/102.8	60.5/101.6	58/236.7	Kent et al. 1995 Kent and Tauxe 2005
Chinle Fm., Upper Red Peak	225	5	49.1/105.1	52.7/107.8	51.4/227.4	Van der Voo and Grubbs 1977
Newark Basin [Princeton core] NJ	227	2.6	52.3/105.3	54.2/106.6	52.8/228.8	Kent et al. 1995 Kent and Tauxe 2005
Chugwater Fm., WY	230	3.3	46.6/113.5	49.8/116.9	45/229.1	Shive et al. 1984
Moenkopi drill core, CO*	230	2.5	51.5/117.4 [R] 56/101.3 [O]	55.9/121.1	46.8/238.3	Baag and Helsley 1974
Upper Moenkopi Fm., CO*	230	3.1	50.1/118.7 [R] 54.8/103.3 [O]	53.9/122.5	45/236.6	Helsley and Steiner 1974
Upper Moenkopi Fm., AZ*	230	4.5	54.7/112.6 [R] 58.5/94.8 [O]	57.5/115.5	50.2/237.5	Steiner et al. 1993
Moenkopi Fm., NM*	234	3.4	53.2/117.1 [R] 57.6/100.3 [O]	55.7/118.8	47.7/236.9	Molina-Garza et al. 1991
Moenkopi Fm., Gray Mt., AZ*	234	5	52.7/113.9 [R] 56.7/97.1 [O]	55.6/116	48.9/235.5	Purucker et al. 1980
Moenkopi Fm., NM*	234	4.9	50.6/124.1 [R] 56/109 [O]	58/119.6	48.7/240	Molina-Garza et al. 1991
Mean of the North American Craton [$N = 17$ poles]	Late Trias.	2.6			52.4/234	Present authors

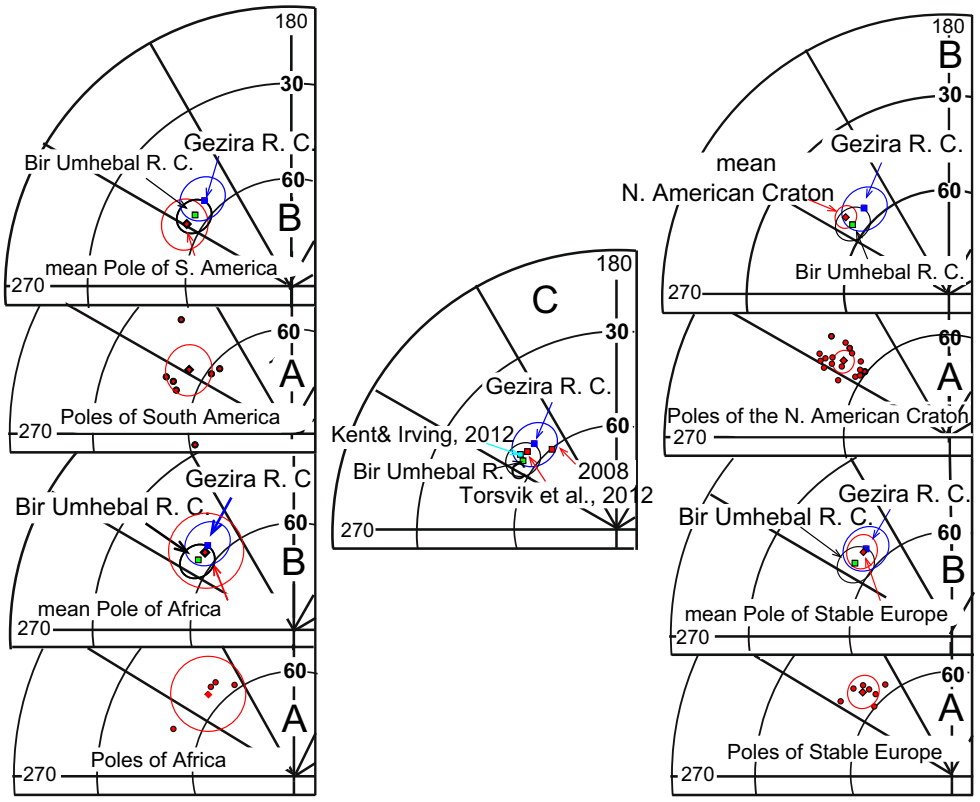


Fig. 6 **a** The Late Triassic Paleomagnetic north poles of the main tectonic units rotated to NE Africa coordinates (Tables 3, 4, 5 and 6) and their respective mean Late Triassic poles (Table 8). **b** The mean rotated poles of the four main continents compared to the Late Triassic poles of the present study with their semi-angles of the cones of 95% confidence [α95] (Table 8). **c** Poles of the present study with their [A95] compared to the mean late Triassic [220–225–230 Ma] poles from the recent global APWPs (Torsvik et al. 2008, 2012) rotated to NE Africa coordinates

(Table 1). Comparing this hematite component with younger poles from Egypt, it resembles the Middle-Late Cretaceous poles of alkali olivine basalt [104 ± 7 Ma] and the younger Trachyte/Phonolite ring dykes [86–78 Ma] of the alkaline volcanic field of Wadi Natash, which yield two paleomagnetic poles at (55°N/250°E) and (66.5°N/229°E), respectively (Lotfy 2011). Accordingly, the hematite anchored was considered as a possible chemical remagnetization carried by secondary hematite, during the nearby splash of Wadi Natash volcanic field in Southeastern Desert during the Late Cretaceous (104–78 Ma), and therefore, will not be further considered.

On the other hand, the coherent magnetite component was consistently recognized in ten sites in Gezira RC and eleven sites in Bir-Umhebal RC. This enhanced its consideration as the characteristic remanent magnetic direction [ChRM] of both complexes, representing the magnetic field during their emplacement in the Late Triassic.

In order to constrain the stability and reliability of the magnetite component characterizing both rings, the reversal test of Mcfadden and Lowes (1981) was applied. In the Gezira RC, the rock-unit mean direction of the magnetite component was (352.6°/−23.5°). The normal polarity of the magnetite component was recorded in three sites (Dec./Inc. = 352.2°/−24.5°) with

Table 4 The Late Triassic Paleomagnetic North poles of Stable Europe rotated to NE Africa coordinates using Euler pole rotations parameters (Table 2). N.A.C. denotes poles in the North American Craton coordinates

Rock unit	Age	K/α95°	North Pole °N/°E	NE Africa °N/°E	Reference
Mercia Mudstone Somerset, UK	217 ± 7	31/5	50/128	54.5/227.5	Briden and Daniels 1999
Sunnhordland Dykes, Norway	223 ± 6	/4.6	50/125	56/225.5	Walderhaug 1993
Taimyr sills, Siberia, Russia	228 ± 1	/2.9	47/122	57/220.5	Walderhaug et al. 2005
Gipskeuper Seds, W Germany	232 ± 3	209/6	49/131	52.5/226.5	Edel and Duringer 1997
Heming LS, Paris Basin, Fr.	238 ± 3	97/3.4	54/141	49.5/238	Theveniaut et al. 1992
Musselkalk F, Poland	238 ± 3	40/12	53/123	59/231	Symons et al. 1995
Frauenbach and Ficoden seria, Thyringia, Germany	233 ± 25	27/5	50.5/135	51/230.5	Bucha et al. 1976
Mean of Stable Europe poles [*N* = 7 poles]	L. Triassic	287/3.6		54.3/228.7	Present study

Table 5 The Late Triassic paleomagnetic North poles of South America (Amazonia and Colorado) rotated to NE Africa coordinates using Euler pole rotations parameters (Table 2)

Rock unit	Age	K/ α95°	North Pole °N/°E	NE Africa °N/°E	Reference
LosColorados, Mendoza, Arg	216±	/7.6	82/118	51.5/245.5	Vizán et al. 2004
Cacheuta Gp Volcs, Arg.	221 ± 14	36/13	74/86	59.5/231	Valencio 1969
Bandui Gp, Br.	225 ± 25	131/2.3	64/137	43/223.5	D'Argella-Filho et al. 2000
Mondoza Lava, Arg.	227 ± 19	−/8	72/88	59.5/229	Creer et al. 1970
Volcs Ischichua basin, Arg.	229 ± 5	14/14	81.7/42	59.5/−103	Valencio et al. 1975
Dolerite Dykes, Suriname	232 ± 5	29/10	82/140	52/246.5	Veldkamp et al. 1971
Mendoza Lavas S Nihuil, Arg	243 ± 2	/4	81/102	54.5/244	Creer et al. 1970
Mean of South America [N = 7 poles	L.Triassic	79/6.9		54.7/239.5	Present study

observed length of the vector resultant $R = 2.94$ and a precision parameter K = 35.48. On the other hand, the reversed polarity was recorded in seven sites (Dec./Inc. = 172.7°/23.1°) with R = 6.81 and K = 32.27. In the Bir-Umhebal complex, the rock-unit mean direction of the magnetite component was (347.7°/−14.9°). The normal polarity of the magnetite component was recorded in five sites (Dec./Inc. = 343.5°/−13.9°) with observed length of the vector resultant $R = 4.96$ and a precision parameter $K = 114$. On the other hand, the reversed polarity was recorded in six sites (Dec./Inc. = 171.3°/15.7°) with $R = 5.89$ and $K = 46.38$. In both complexes, the normal and reverse polarity sites of the magnetite anchored component share a common mean at the 95% confidence, and then, positively pass the reversal test of McFadden and Lowes (1981). This implies that the antipodal means of the magnetite component, in both complexes, are free from disturbing secondary overprints.

Subsequently, for each ring complex an overall mean direction of the PDF overprint together with the diagnostic ChRM direction and its corresponding mean paleomagnetic pole were calculated with their associated statistical parameters (Fisher 1953), tabulated (Table 1) and plotted in Figs. 4 and 5.

Discussion and Euler pole rotation

In order to probe the fidelity of the Late Triassic paleomagnetic poles obtained from the Gezira and Bir-Umhebal ring complexes, the high-quality Late Triassic poles representing the main tectonic units, were selected from the world paleomagnetic databases (Van der Voo 1993; McElhinny and Lock 1996; Pisarevsky 2005) and various comprehensive literatures (Torsvik et al. 2008, 2012). These global Late Triassic poles were, then, rotated from their tectonic plates to the Northeast African coordinates using the appropriate Euler pole rotation parameters (Table 2).

As one of the best studied continents, seventeen paleomagnetic poles were selected from the North American Craton. Poles of the Colorado Plateau were, first, internally rotated to the Craton using the rotation parameters of Beck and Housen (2003), These Colorado Plateau poles with the North American Craton poles were, then, corrected for inclination shallowing errors to remove the influence of the inclination shallowing in sedimentary Triassic red beds on the North American pole positions. The inclination-corrected North American poles are, then, rotated to NW African coordinates (Labails et al. 2010), followed by internal rotation to NE Africa (Torsvik et al. 2012). The rotated North American Craton poles show high internal consistency (Table 3, Fig. 6) and yield a Late Triassic mean pole in the NE African coordinates at 52.4°N/234°E [K = 196.6/α95 = 2.6°] (Tables 3, 4). This North American mean pole lies very close to the poles of the Gezira and Bir-Umhebal complexes, with its semi-angle of the cone of 95% confidence lying within that of the Gezira ring and partially overlap with that of Bir Umhebal ring (Table 7, Fig. 6).

Table 6 The Late Triassic paleomagnetic North poles of Africa rotated to NE Africa coordinates using Euler pole rotations parameters (Table 2)

Rock unit	Age	K/α95°	North Pole °N/°E	NE Africa °N/°E	Reference
Argana red beds, Morocco	227 ± 19	7.5/12	50.6/251.4	48.5/254	Martin et al. 1978
Up, Tr. Sed., Tunisia	227 ± 19	37/11.5	55/223.3	55/223.3	Ghorabi and Henry 1991
Al Azizia F., Libya	230 ± 19	36/5.5	54.5/225.8	54.5/225.8	Muttoni et al. 2001
Al Azizia F., Libya	230 ± 5	36/4	59.3/214	59.3/214	Muttoni et al. 2001
Mean of Africa [N = 4 poles]	L.Triassic	53/12.7		55.3/230.3	Present study

Table 7 A. The middle Late Triassic Paleomagnetic poles of the African Plate of the present study. B. The Mean Late Triassic poles rotated from the main tectonic units with the overall mean of four main continents rotated to NE Africa coordinates. C. Mean Late Triassic [230–220 Ma] mean poles from recently calculated global apparent polar wander paths [APWPs] rotated to NE Africa coordinates. [K/α95] denote the precision parameter and the semi-angle of the cone of 95% confidence around the mean pole. Cairo Ref. and Plat. denote the corresponding Late Triassic paleo-declination, paleo-inclination and paleo-latitude predicted at the reference location of Cairo, Egypt [30°N/31°E] from the cited poles

A	Ring Complex	Age [Ma]	Sites	North Pole [NE Africa]	K/α95°	Cairo Reference 30°N/31°N	Plat.
1	Gezira R. C. [22.3°N/33.6°E]	229 ± 5 Ma [K/Ar biot.]	10	54.6 N/226.2E	5/9	351.2/−8.7	−4.4
2	Bir Umhebal R. C. [23.75°N/33.3°E]	223 ± 9 Ma [Rb/Sr]	11	56.4 N/236E	3/6	346.5/−2	−1
3	Mean of [1 + 2] [23°N/33.5°E]	229–223 Ma [L. Trias.]	21	56 N/231E	3/6	349/−4.6	−2.3
B	Tectonic Unit	Age [Ma]	N	Mean N, pole [NE Africa]	K/A95°		
1	North American Craton	226 ± 10	17	52.4/234	197/2.6	346/−10.2	−5.2
2	Stable Europe	226 ± 10	7	54.3/228.7	287/3.6	350/−8.6	−4.3
3	S. America (Amazonia and Colorado)	226 ± 10	7	54.7/239.5	79/6.9	344/−3.6	−1.8
4	Africa	226 ± 10	4	55.3/230.3	53/12.7	349/−6	−3
	Mean of four continents	226 ± 10	4	54.2/233.1	703/3.5	347.3/−7.3	−3.7
C	Global APWP	Age [Ma]	N	Mean N pole [NE Africa]	A95°		
1	Global APWP (Torsvik et al. 2008)) rotated to NE African coordinate	220	29	62/217.5	2	357/4	2
		230	28	58/220.5	2.6	355/−3.3	−1.7
		Mean [225]		60/219		356/.5	0.3
2	Global APWP (Torsvik et al. 2012)) rotated to NE African coordinate	220	35	58/230.5	2.2	350/−1	−0.5
		230	33	52.5/230	2.7	348.5/−11.5	−5.8
		Mean [225]		55.2/230.2		349/−6.5	−3.3
3	APWP (Kent and Irving 2010), rotated to NE African coordinate	220	8	58.9/233.5	2.3	349/1.5	0.8
		230	3	56.3/232.5	5.7	348/−3.6	−1.8
		Mean [225]		57.6/233		348.4/−1	−0.5

Seven Late Triassic poles from Stable Europe (Table 4) were chosen and rotated to the North American Craton using the rotation parameters of Alvey (2009), then, followed the rotation sequence of the American Craton poles to the NE African coordinates. The rotated poles are reasonably clustered about a mean pole at 54.3°N/228.7°E [$K = 287/\alpha95 = 3.6°$] (Tables 4 and 7, Fig. 6). The Late Triassic mean rotated pole of Stable Europe lies midway between the Gezira and Bir-Umhebal poles with their semi-angle of cones of 95% confidence almost completely overlapping (Fig. 6).

From both Amazonia and Colorado, seven Late Triassic poles were selected from South America (Table 5). These poles were, first, rotated to South Africa using separate rotation parameters for Amazonia and Colorado (Torsvik et al. 2009), then internally rotated to NE Africa (Torsvik et al. 2012). The rotated poles show some dispersion with their mean pole at 54.7°N/239.5°E [$K = 79/\alpha95 = 6.9°$) (Table 5, Fig. 6). The South American mean pole in the far side of Bir-Umhebal along the same latitude but at higher longitude, with its semi-angle of the cone of 95% confidence enclosing that of Bir-Umhebal pole (Fig. 6).

From Africa only four Late Triassic poles from Northwest Africa were collected and internally rotated to NE Africa using the rotation parameters (Torsvik et al. 2012). These poles are dispersed (Table 6, Fig. 6) yielding a mean pole at 55.3°N/230.3°E [$K = 53/\alpha95 = 12.7°$]. Despite the dispersion of the African poles, the mean pole lies between Gezira and Bir-Umhebal poles, with its wide semi-angle of the cone of 95% (12.7°) enclosing their cones (Fig. 6).

A mean pole of 225 Ma was calculated for the Late Triassic poles [230 and 220 Ma] of the recently published global APWPs (Torsvik et al. 2008, 2012). These poles were, then, internally rotated to NE African coordinates (Torsvik et al. 2012) for comparison with the Gezira and bir-Umhebal poles (Table 7, Fig. 6c). The three poles of Torsvik et al. (2012) form a path between our poles. On the other hand, the three poles of Torsvik et al. (2008) lie in the far side of the Gezira pole at higher latitude and lower longitude than both our poles and those of Torsvik et al. (2012) (Table 7, Fig. 6c). Although that the reason of the deviation between the global APWPs of Torsvik et al. (2008, 2012) is not clear, it can be referred to the new and possibly more accurate rotation parameters used in Torsvik et al. (2012).

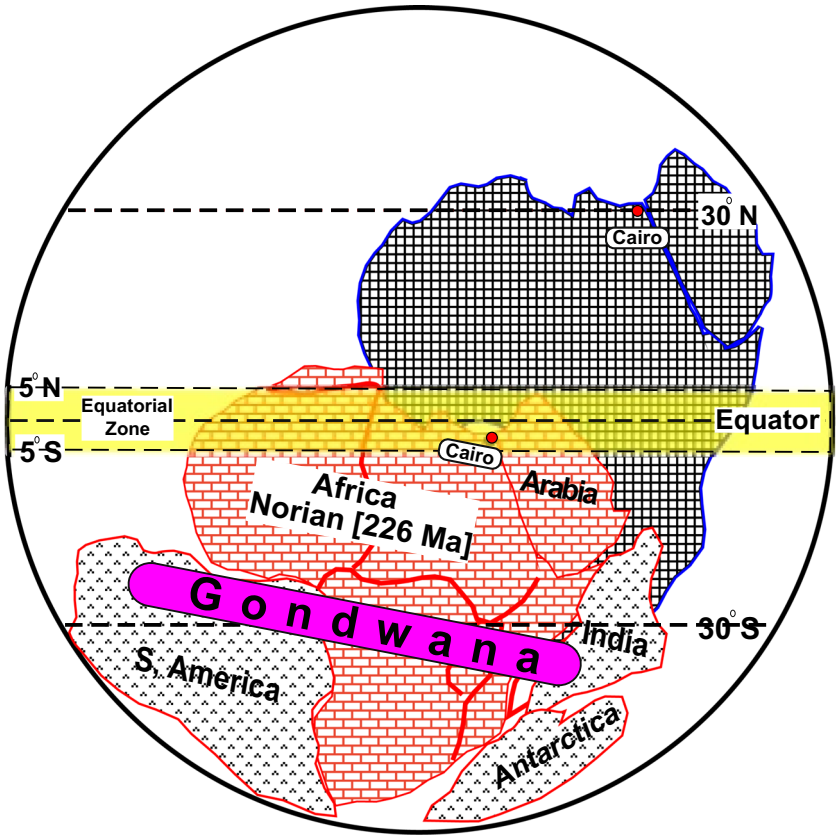


Fig. 7 Paleotectonic position of Africa within Gondwana in the Late Triassic showing Northeast Africa (Cairo) south the Equator but within the equatorial zone [±5° around the Equator]. Present-day position of Africa with Cairo at 30°N is presented for comparison

Conclusion and the Late Triassic paleotectonic position of North Africa

The reasonable consistency of the Late Triassic [Carnian-Norian] poles of the Gezira and Bir-Umhebal ring complexes with their synchronous poles rotated from the well-studied North American Craton and Stable Europe (Table 7, Fig. 6), greatly constrained their reliability as representative of the geomagnetic during the emplacement of the two complexes. In order to infer the paleotectonic position of the African Plate in the Late Triassic from the paleomagnetic poles of the Gezira (229 ± 5 Ma) and Bir-Umhebal (223 ± 9 Ma) complexes, the predicted paleo-

Table 8 The middle Late Triassic paleomagnetic poles of the Africa Plate of the present study (Upper part) compared to the coeval poles of some North Mediterranean tectonic blocks (lower part) with their predicted paleo-latitudes calculated at the reference location of Cairo (Egypt) [30°N/31°E]

	Tectonic Unit	Age [Ma]	K/α95°	Paleomagnetic Pole [°N/°E]	Paleo-lat. at Cairo	Reference
1- Gezira R.C.	African Plate	229 ± 5		54.6/226	−4.4	Present study
2- Bir Umhebal RC	African Plate	223 ± 9		56.4/236	−1	Present study
Mean of 1 + 2	African Plate	226 ± 3		56/231	−2.3	Present study
Pleader red beds, Sp.	Iberian Meseta	221 ± 13	47/3	63/178	6.5	Van der Voo 1967
Catalunian Red beds, NE Sp.	Iberian Meseta	235 ± 5	205/6	51/181	−5	Pares et al. 1988
Algarve Red beds	Iberian Meseta	235 ± 5	33/11	62/202	2.5	Pares and Dinares-Turell 1994
Meseta Red beds	Iberian Meseta	235 ± 5	24/9	54/178	−1.5	Pares and Dinares-Turell 1994
Garldo Red beds, Koram	MesetaTerrain Pyrenees	226 ± 18	−/−	55/196	−4	Van der Voo 1969
N Murag-M, Sardinia	Corsica/Sardinia	235 ± 5	999/4	32/285	3.5	Horner and Lowrie 1981
Reifling Fm, Mendlinbach Sect.	Calc. Alps	235 ± 2	32/2.7	59/210	−1	Gallet et al. 1998
Pelvaux Spilite, Fr.	Alps N. & W of Inspuric line	215 ± 10	13/27	55/235	−2.5	Westphal 1973
Volc Sabb SS, It.	S Alps, Autoch. Italy	231 ± 4	14/13	57/215	−3	Muttoni et al. 1996
Ladinean/Carn Volcs	S Alps, Autoch. Italy	230 ± 3	22/9	48/240	−7.5	Manzoni 1970
Buchenstein beds	N. Italy	237 ± 3	39/20	61/217	1	Brack and Muttoni 2000
AtlantisVolcs, Pelagonis, Greece	Greece, S Bulgaria	230 ± 15	44/9	54/221	−6	Turnell 1988
Anisian/Ladinian Boundary Section	Hydra Isl., Greece	239 ± 2	20/28	38/221	−21	Muttoni et al. 1997
Saharonim Fm, Ramon anticline	Israel	227 ± 19	−/2.4	44/254	−6	Freund and Tarling 1979

declinations, paleo-inclinations and paleo-latitudes at Cairo (30°N/31°E) were calculated (Table 7). Subsequently, it can be concluded that:

1. In terms of paleo-latitude, during the Late Triassic [229–223 Ma], the African Plate was about 34°–31° of latitude south to its present-day latitude. Cairo which is now at 30° N was at 4° S during the emplacement of the Gezira complex (229 ± 5 Ma) and at 1° S during the formation of the Bir-Umhebal (223 ± 9 Ma). The average Late Triassic (229–223 Ma) paleo-latitude of Cairo was 2.3° S (Table 7, Fig. 7). This means that the Cairo in NE Africa was moving northwards, in the equatorial region, just south of the Equator from 4° S to 1° S between 229 Ma and 223 Ma.
2. In terms of paleo-azimuth, the African Plate was 13°–9° NNE with respect to the present-day [N-S] axial geocentric dipole (Table 7, Fig. 7).

In order to, further, constrain our equatorial paleo-latitudes of NE Africa in the Late Triassic from the Gezira and Bir-Umhebal poles, their predicted paleo-latitudes at the reference location of Cairo were, then, compared with:

1- The predicted paleo-latitudes of the mean Late Triassic poles rotated from the main tectonic Plates and the co-eval poles rotated from the Global APWPs (Torsvik et al. 2008, 2012) (Table 7). The inspection of the last column containing the predicted paleo-latitudes of Cairo constrains its Late Triassic equatorial location.
2- The predicted paleo-latitudes of the main Mediterranean tectonic blocks such as The Iberian Meseta, Corsica/Sardinia, Italy, Sicily, Greece and Israel (Table 8) are mostly equatorial. However, the comprehensive glance on last column in Table 8, unequivocally, support the equatorial paleo-latitude of NE Africa during the Late Triassic [229–223 Ma] with all the Mediterrannean blocks nearby in the equatorial region.

Acknowledgments This research would not have been possible without the kind support and help of Prof. Dr. Hatem H. Odah, President of the National Research Institute of Astronomy and Geophysics [NRIAG]. The authors are deeply indebted to all colleagues and staff members in the seismology Center of Aswan (NRIAG) for their tremendous helps and support during the field work. We wish, also, to express our sincere gratitude to all staff members and researchers in the Geomagnetism Laboratory, for their persistent efforts during the progress of this research. This research was supported by the official research funds of Menia University given to the Faculty members at the Faculty of Science for the first author and special fund and logistics offered by the National Research Institute of Astronomy and Geophysics [NRIAG] to the second author. Sincere thanks are due the reviewers of this manuscript, whose comments greatly improve the final version of this research.

The authors are, also, sincerely grateful to the anonymous reviewers of this research for their comprehensive comments which significantly improved our research.

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